

Module3_Lab2

September 17, 2026

```
[1]: import pandas as pd
df = pd.read_csv("usarrests.csv")
df.head()
```

```
[1]: Unnamed: 0  Murder  Assault  UrbanPop
0     Alabama    13.2    236.0         58
1     Alaska     10.0    263.0         48
2     Arizona     8.1    294.0         80
3     Arkansas     8.8    190.0         50
4  California     9.0    276.0         91
```

```
[2]: df.shape()
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[2], line 1
----> 1 df.shape()

TypeError: 'tuple' object is not callable
```

```
[3]: df.shape
```

```
[3]: (50, 4)
```

```
[4]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 50 entries, 0 to 49
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Unnamed: 0   50 non-null    str
1   Murder       50 non-null    float64
2   Assault      49 non-null    float64
3   UrbanPop     50 non-null    int64
dtypes: float64(2), int64(1), str(1)
memory usage: 2.1 KB
```

```
[5]: df.describe()
```

```
[5]:      Murder      Assault  UrbanPop
count  50.00000  49.000000  50.00000
mean    7.78800  182.183673  74.20000
std     4.35551  130.877435  73.40828
min     0.80000   45.000000   6.00000
25%     4.07500  109.000000  53.25000
50%     7.25000  159.000000  66.00000
75%    11.25000  249.000000  77.75000
max    17.40000  879.000000  570.00000
```

Fist, I ran the four commands on every new data set.

```
[6]: df.isnull().sum()
```

```
[6]: Unnamed: 0      0
Murder          0
Assault         1
UrbanPop        0
dtype: int64
```

There is one state with missing value.

```
[7]: df[df["UrbanPop"] > 100]
```

```
[7]:      Unnamed: 0  Murder  Assault  UrbanPop
14      Iowa      2.2    56.0      570
```

This one is strange because it shows that Iowa has an Urban Population % of 570, which exceeds 100%.

```
[ ]: df[df["Murder"] < 1]
```

The 0.8 Murder rate for North Dakota is a strange one, too, at 0.8